Q3. Article Summarization into 50 Words

Aim: To design three prompting methods to summarize an article into 50 words and compare the generated summaries based on accuracy, completeness, and readability.

Algorithm

1. Start the program.
2. Read the given article.
3. Create a Zero-shot prompt to summarize the article in 50 words.
4. Create a One-shot prompt with one example of article summarization.
5. Create a Few-shot prompt with multiple summarization examples.
6. Generate a summary using each prompt.
7. Count the words in each summary.
8. Compare the summaries based on accuracy, completeness, and readability.
9. Display the summaries and comparison results.
10. Stop the program.

Article

Title: Artificial Intelligence in Healthcare

Artificial Intelligence is increasingly used in healthcare. It helps doctors analyze medical images, predict health risks, support personalized treatment, and manage hospital workflows. AI-powered wearable devices monitor activity and vital signs. However, healthcare AI faces challenges such as data privacy, biased datasets, and inaccurate predictions. Human supervision and careful testing are essential for safe and responsible use.

1. Zero-shot Prompt

Prompt:

Summarize the following article in exactly 50 words. Preserve the main ideas, important applications, benefits, and challenges. Do not introduce information that is absent from the article.

Article: [Insert the article provided above]

Output — 50 words:

Artificial Intelligence supports healthcare through medical image analysis, health-risk prediction, personalized treatment, hospital workflow management, and wearable monitoring. These

applications can improve medical decision-making and patient care. However, privacy concerns, biased datasets, and inaccurate predictions remain challenges. Careful testing and human supervision are essential for safe and responsible healthcare AI implementation.

2. One-shot Prompt

Prompt:

Example:

Article: Solar energy produces electricity from sunlight. It is renewable and can reduce dependence on fossil fuels. However, installation costs and weather conditions can affect its use.

Summary: Solar energy generates renewable electricity, reduces reliance on fossil fuels, and offers environmental benefits. However, installation expenses and weather conditions can influence its effectiveness.

Now summarize the healthcare article in exactly 50 words. Preserve its main points and use clear English.

Article: [Insert the article provided above]

Output — 50 words:

AI is improving healthcare by assisting medical image analysis, predicting health risks, supporting personalized treatment, managing hospital workflows, and monitoring vital signs. Despite these benefits, privacy concerns, biased datasets, and inaccurate predictions remain important challenges. Careful testing and human supervision are necessary to ensure that AI is used safely and responsibly in healthcare.

3. Few-shot Prompt

Prompt:

Example 1:

Article: Online learning provides flexible access to educational resources. Students can learn remotely, but internet access and reduced face-to-face interaction may create challenges.

Summary: Online learning offers flexible remote education but may be limited by internet access and reduced face-to-face interaction.

Example 2:

Article: Electric vehicles reduce direct exhaust emissions and can lower running costs. However, charging infrastructure, battery production, and purchase prices remain concerns.

Summary: Electric vehicles can reduce exhaust emissions and running costs, although charging access, battery production, and purchase prices remain challenges.

Now summarize the healthcare article in exactly 50 words. Include the major applications, benefits, challenges, and need for human supervision.

Article: [Insert the article provided above]

Output — 50 words:

Healthcare AI supports medical imaging, health-risk prediction, personalized treatment, hospital management, and wearable monitoring. It can improve healthcare decision-making and patient services. However, privacy risks, biased datasets, and inaccurate predictions require attention. Careful testing and human supervision are essential to ensure that AI systems are used safely and responsibly in healthcare settings.

Comparison of summaries

Criteria	Zero-shot	One-shot	Few-shot
Accuracy	High	High	High
Completeness	Covers major points	Covers major points	Covers major points
Readability	Clear	Clear and concise	Clear and structured
Examples provided	None	One	Two

Result: All three summaries preserve the article's key ideas. One-shot and few-shot prompting provide examples that guide the summarization style.